DAY 01 · THEORY HANDOUT

# AI, ML and DL: First Steps

What the names mean, how learning works, and where we will run our notebooks.

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# AI, ML, and DL: three linked names



**Artificial Intelligence** is the broad field. ML is inside AI. DL is inside ML.




A fixed set of rules can help a computer choose an answer. **Machine learning** instead learns useful patterns from examples.

**Deep learning** uses neural networks with many connected layers. A layer is one small processing step. "Deep" refers to the layers.

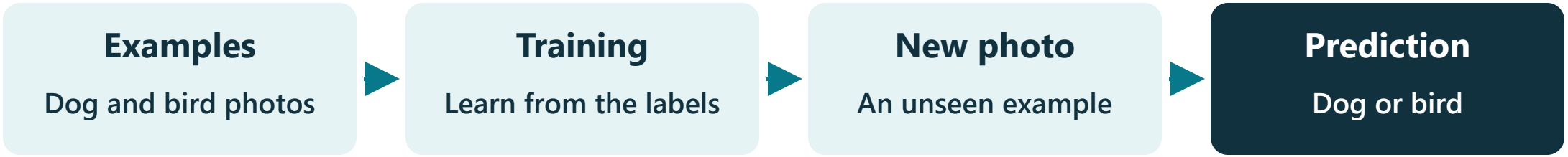
> These are related names, not three separate kinds of computer.

# How does a model learn?



**Data** means the examples we give it. A **label** is the known answer for one example.

**A label is the known answer: dog or bird.**




We check new photos to see whether the learned model works.

**Training** is the learning stage. The program changes its internal numbers to find patterns that help match photos to their labels.

A **model** is the system with those learned numbers. A **prediction** is its answer for a new example. It can be wrong.

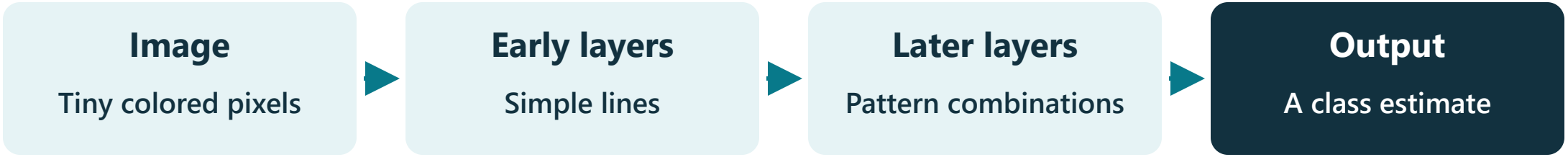
> Learning from known answers is called supervised learning. Testing uses examples that were not used for learning.

# What makes learning deep?



A **neural network** is a model made of connected processing layers.

## Neural network = connected layers of small calculations




Early layers can find simple patterns. Later layers can combine them. The model learns which patterns help with the task.

More layers can help with hard patterns, but they often need more data and computer work. They do not give a computer human thought.

> DL is one kind of ML. A small ML model can still be a good choice for a simple task.

# GenAI, prompts, and language models



**Generative AI** makes new content, such as text, pictures, or sound.

### Prompt

The instruction we give:
"Explain a rainbow."

### LLM

Large Language Model:
a model trained on text.

### AI agent

Uses a model and tools
to carry out task steps.

An LLM creates text one small piece at a time. A piece may be a word or part of a word. It learns patterns from many text examples.

An agent can use tools, such as search or a calculator. We set what it may do and check its results.

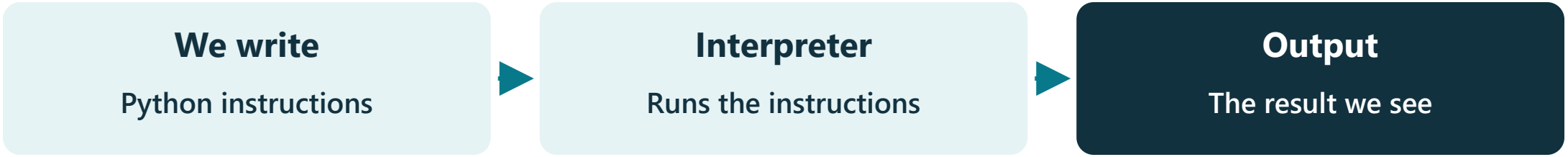
> A clear reply can still contain a false fact. Check important answers before using them.

# Python: our way to give instructions



**Python** is a programming language. A **program** is a set of instructions for a computer.

## Interpreter = software that runs Python




Python is easy to read and has many ready tools for data and AI. A **library** is a collection of code we can reuse.

Python is one choice. C++ often runs fast parts behind the scenes. JavaScript can build web pages. We start with Python to learn the ideas with less code.

> Installing Python gives us the interpreter. We also need a place to write and run our work.

# IDE, Jupyter, and kernel



An **IDE** is an Integrated Development Environment: an app with tools for writing and running code.

### IDE: the workspace

Example: VS Code.

Write and manage files.

### Jupyter: the notebook

Text + code + results.

A file ending in .ipynb.

### Kernel: the runner

Runs the code cells.

Remembers their values.

A **Jupyter Notebook** is a document with separate cells. A Markdown cell holds text or pictures. A code cell holds instructions to run.

The **kernel** is the running Python process behind the notebook. The browser shows the page; the kernel does the calculations.

> We can use Jupyter in a browser for this course. Installing VS Code is optional.

# Set up your computer

Use the setup playlist to prepare for the Python lessons.

## Watch only the first 4 videos.

These videos cover:

- Anaconda installation
- VS Code setup
- Jupyter setup
- Git installation and setup

Scan to open the playlist

**Open the setup playlist**

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